# Review of "Mixed-dimensional models of poroelasticity" by J. Stebel and J. Brežina

The paper considers a mixed-dimensional model of a poroelastic medium containing a single, planar fracture. The system of equations is obtained by averaging the Biot equations across the fracture aperture. After defining the appropriate function spaces, the weak formulation of the problem is presented and it is shown to admit a unique solution through the use of a fixed-stress iterative scheme.

I have several concerns regarding this article. First, it is difficult to identify a main contribution to the field of modeling fractured, poroelastic media that is sufficiently significant. The title implies that this work concerns mixed-dimensional models of poroelasticity when in fact, a single, linear model is presented with very few justifications or underlying modeling assumptions.

The key modeling questions such as contact mechanics, friction relationships, the dependency of the fracture permeability on a variable aperture, and the extendability to fracture networks are not discussed aside from brief mentions in the introduction and the last sentence of the manuscript. However, these are the questions that an interested reader will be looking for and have been considered and rigorously analyzed previously, see e.g. [2, 3], cited below.

The presentation moreover leads me to believe that the model is not based on the linearization of physically reasoned principles, but rather follows from a dimensional reduction of a linear system of equations through a standard averaging procedure. In turn, the validity and relevance of this model are not straightforward and need to be addressed.

In terms of analysis, the main result (Theorem 5.1) is obtained by constructing a fixed-stress iterative scheme and showing that it is a contraction. However, it is not clear that these constructions are necessary. In particular, the text after eq (4.9) reads:
"*If the last term on the left hand side was not present, the identity would yield a priori bounds of the solution. However, this last term cannot be*

*estimated by the other terms and may possibly be a source of uncontrolled "energy". Due to this fact we are unable to prove the global in time existence of weak solution without any further restriction of the model data.*"
Immediately afterwards, this problematic term is eliminated so why do we not obtain the a priori bounds, following the authors' expectation? It raises the question why the proof of Theorem 5.1 does not follow the standard theory of linear evolution equations, since the presented strategy seems quite cumbersome in comparison.

The introduction states that there are advantages to the presented approach for the design of numerical methods. However, a demonstration of this is lacking in the manuscript. It is my opinion that the explicit link between the analysis of the model and the implications to computational methods is a missing feature in this work that would have been appreciated, or even expected, by the readers of M2AN.

In order to warrant publication, I believe the manuscript should expand its analysis to a more involved model of fractured poroelastic media or justify the modeling choices and connect the analysis to a numerical method. Moreover, a more thorough comparison is needed to existing literature (in particular to the more complex models in [2, 10, 11] and [1, 2, 3] below) and the main contributions need to be highlighted. This would require fundamental changes to the aim and scope of this manuscript.

## Further Comments

- The notation is very involved, nearly impenetrable. The reader is expected to keep track of tildes and subscripts on differential operators, as well as square, curly, and jump brackets, sometimes in the same equation, e.g. (4.8c). Please make the notation more accessible. For example, the overline notation does not appear at all in the eventual model.

- Page 1: "*In modelling hydro-geomechanical coupling, the mechanical response of fractures is often neglected.*" This statement needs to be founded with appropriate citations. Many of the referenced works do incorporate the response of fractures.

- Page 1: "*However, phenomena like shear-induced opening, friction or healing may play essential role, raising the need for appropriate mathematical description of fracture mechanics*" I agree with this statement, but these phenomena are not considered in this work.

- Page 1: The original idea of dimension reduction cannot be accredited to [12] since vertical integration was already a well-established technique in reservoir simulation by that time. Moreover, modeling fractures as lower-dimensional manifolds was already done previously by e.g. [4], cited below.

- Page 2: "*To cope with the fact that the mechanical part of the problem is a system of equations, ...*" It is not clear which system of equations this refers to and why this requires special consideration.

- Page 2: The fixed-stress splitting scheme was not proposed, but analyzed, in [15].

- Page 4: Without actually making the generalization to fracture intersections, Remark 3.1 seems to be of little value.

- Section 4: Pressure continuity, which is a modeling assumption, is presented after establishing model (4.8). However, (4.8) plays no role in this work so why is (4.10) not directly presented as the model?

- Page 5: The space $\mathcal{V}$ is defined twice, in the same line. Whenever the space (or its norm) is used, an unnecessary confusion arises because one definition implies a scalar and the other a vector-valued function space. Please redefine one of the spaces.

- Page 8: Strictly speaking, Section 5 does not show well-posedness of the poroelastic problem because one needs to show continuity of the solution on the data and such bounds are not explicitly presented.

- Page 10: There is an inconsistency in notation whenever $\{\{\|\cdot\|\}\}$ appears because $\|\cdot\|^{\oplus}$ is undefined.

- Page 14: The optimality of the splitting parameters are of little relevance to the presented existence and uniqueness results. In the context of a numerical scheme, however, the optimality will need to be verified.

- Appendices: Appendix B is a direct consequence of Appendix C. Do these appendices not show the same calculation twice?

- Appendix B: Existence of $\boldsymbol{u}$ is not yet verified so how can $\partial_t \boldsymbol{u}$ be used as a test function?

- Typographic errors: page 1 "accout", page 7 "the the".

## References


[1] I. Ambartsumyan, E. Khattatov, T. Nguyen, and I. Yotov, *Flow and transport in fractured poroelastic media*, GEM-International Journal on Geomathematics, 10 (2019), p. 11.

[2] V. Girault, M. F. Wheeler, B. Ganis, and M. E. Mear, *A lubrication fracture model in a poro-elastic medium*, Mathematical Models and Methods in Applied Sciences, 25 (2015), pp. 587–645.

[3] V. Girault, M. F. Wheeler, K. Kumar, and G. Singh, *Mixed formulation of a linearized lubrication fracture model in a poro-elastic medium*, in Contributions to Partial Differential Equations and Applications, Springer, 2019, pp. 171–219.

[4] M. Karimi-Fard, L. J. Durlofsky, K. Aziz, et al., *An efficient discrete fracture model applicable for general purpose reservoir simulators*, in SPE Reservoir Simulation Symposium, Society of Petroleum Engineers, 2003.